

BSCS FINAL YEAR PROJECT PROPOSAL

PLAYERZWZ

Term of Registration: Spring 2026



Particulars of the students:

Sr. #	Registration#	Name in Full Use Block Letters	Contact #	CGPA
1	L1F24PACS0021	WALEED ABID	0316 4708028	2.4
2	L1F21BSCS0951	AFAQ UL ISLAM	0321 7080500	2.92
3	L1F22BSCS1053	HASSAN AHMED	0307 6271232	2.06

Project Type:

- | | | |
|---|---|--|
| ☐ Research based | ☐ Hardware based/Embedded | ☐ Game based |
| ☐ Software Development | ☐ Artificial Intelligence (AI) | ☒ Mobile Application |
| ☐ Web Application | ☐ Robotics | ☐ Database |
| ☐ Other: _____ | | |

Faculty of Information Technology &
Computer Science

University of Central Punjab

Project Title

PLAYERZWZ

Project Advisor

Mr. Asif Farooq

Acceptance of Project Idea

Students will be required to defend their idea before the scrutiny committee (SC) which has the authority to accept or reject the project idea. The decision taken by the SC will be final and cannot be challenged.

Similarly, in phase wise evaluations, the marks awarded by the evaluators will be considered final. No excuses on their skill, relevancy, competency or biasedness will be acceptable as an absolve.

Supervisory Meetings

A group is required to hold at least two meetings per month and maintain meeting minutes. Missing two consecutive meetings without any notice may lead to withdrawal of the project.

Advisor's Consent

I, Mr. Asif Farooq, am willing to guide these students in all phases of above-mentioned project as advisor. I have carefully seen the Title and description of the project and believe that it is of an appropriate difficulty level for the number of students named above.

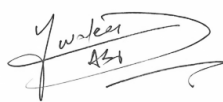
Note:

Advisor can't be changed and the duration for completion of the Project is 2 regular semesters (approx.) from the date of Registration of the Project.

Signatures and Date

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Advisor

Sr. #	Name	Registration #	Signatures
1	WALEED ABID	L1F24PACS0021	
2	AFAQ UL ISLAM	L1F21BSCS0951	
3	HASSAN AHMED	L1F22BSCS1053	

1. Abstract

The Playing sports regularly is important for both physical health and mental well-being, yet many people are unable to do so because they struggle to find players at the right time. Very often, games are cancelled simply because one or two players are missing, friends are busy, or coordination becomes difficult at the last moment. In addition, booking sports grounds or courts is usually a manual and time-consuming process, involving phone calls, messages, or physical visits.

PLAYERZWZ is proposed as a mobile application that aims to solve these everyday problems. The application allows users to find nearby players, create or join sports games, and book sports arenas based on their location, availability, and preferences. Artificial Intelligence is used to suggest suitable matches, recommend reliable players, and reduce the chances of no shows.

The idea of this project comes from real life experiences of students and sports enthusiasts who frequently face difficulties in organizing games. The expected outcome of PLAYERZWZ is a practical, user-friendly platform that minimizes waiting time, improves coordination, and encourages people to stay active through sports

2. Introduction of the Project

PLAYERZWZ is a mobile based sports matchmaking and arena booking application designed to make sports planning easier and faster. Instead of relying on personal contacts or informal groups, users can use PLAYERZWZ to instantly find nearby players who are interested in the same sport and are available at the same time.

The application uses location-based services to show nearby players and sports facilities, while Artificial Intelligence helps in recommending suitable games and players based on previous activity and behavior. The project lies in the domain of mobile application development, artificial intelligence, and location-based systems. With busy academic and professional schedules, people often find it difficult to organize sports activities. PLAYERZWZ addresses this issue by providing a simple and intelligent solution that reduces coordination effort and makes sports participation more accessible.

3. Problem Statement & Gap Analysis

Many people want to play sports such as cricket, football, badminton, or squash, but face difficulties in finding players on short notice. Common problems include incomplete teams, last minute cancellations, unreliable participants, and lack of proper coordination. Currently, most players depend on WhatsApp groups, phone calls, or personal networks, which are limited and inefficient. Existing applications either focus only on sports venue booking or provide general social networking features. There is no integrated platform that intelligently connects players, evaluates reliability, and supports real time matchmaking.

The main gaps identified are:

- Difficulty in finding nearby players instantly
- No intelligent system for recommending suitable matches
- Lack of trust when playing with unknown players
- Manual and fragmented arena booking methods

In addition to player-related challenges, a market survey of local sports complexes and arenas will be conducted to understand current booking methods, availability issues, pricing structures, and operational gaps faced by facility managers. PLAYERZWZ is designed to bridge these gaps by offering AI-assisted matchmaking, reputation scoring, and location-based discovery in a single application.

4. Proposed Solution / Product Concept

PLAYERZWZ provides a simple yet effective solution for organizing sports activities. Users can create a game by selecting the sport, location, date, time, and number of players required. Nearby users can view these games and join them if they are interested and available. The system uses Artificial Intelligence to recommend games and players based on factors such as past participation, preferences, availability, and reliability. A rating and reputation system helps users identify dependable players and discourages repeated no shows.

Key features include:

- Location based player discovery
- Game creation and joining
- AI assisted match recommendations

- Sports arena discovery and booking
- Player rating and reliability system

This approach reduces manual effort, saves time, and improves the overall sports experience.

5. Objectives & Target Market

The main objective of PLAYERZWZ is to develop a reliable and intelligent platform that enables quick sports matchmaking and efficient arena booking. The project aims to reduce game cancellations, improve player coordination, and promote regular sports participation. The target users include students, young professionals, and sports enthusiasts aged between 16 and 45 who actively participate in sports and prefer convenient digital solutions.

6. Customer & User Research

Informal discussions and surveys conducted among students and local sports players revealed that many games are cancelled due to missing players. Participants shared that they often waste time coordinating through messages and calls. A common concern was the lack of trust when playing with unknown people.

Users expressed interest in an application that could quickly connect them with nearby players and highlight reliable participants. These insights played a major role in shaping the features of PLAYERZWZ, particularly the AI based recommendation and reputation system.

7. Tools and Technologies

- Frontend: Flutter (Dart)
- Backend: Python (Flask)
- Database: Firebase Firestore
- Map & Location: Google Map API
- AI/ML: Scikit-learn, Panda ,Numpy

8. Expected Outcomes /KPI's

The expected outcomes of PLAYERZWZ include faster player matching, fewer cancelled games, improved trust among users, and increased participation in sports

activities. From an academic perspective, the project provides hands on experience in mobile development, backend systems, and applied artificial intelligence.

9. Completeness Criteria

The PLAYERZWZ project will be considered complete when all core features of the system are fully implemented, tested, and functioning as intended. The application should allow users to register, create profiles, discover nearby players, create and join sports games, and receive AI-assisted recommendations without system failure.

From a technical perspective, completeness will be achieved when the mobile application, backend services, database, and AI components are properly integrated. The matchmaking logic, reputation scoring, and notification system must work reliably under normal usage conditions. Basic security measures such as authentication, data validation, and user reporting must also be operational.

In addition, proper documentation, testing results, and a working prototype demonstration will be required to mark the project as complete. The final submission must reflect a stable, usable system rather than a partial or conceptual solution.

10. Challenges

Several challenges are expected during the development of PLAYERZWZ. One major challenge is managing real-time location-based matching while ensuring acceptable performance and accuracy. Handling varying user availability and preventing false or outdated matches will require careful logic design.

Another challenge involves implementing Artificial Intelligence in a meaningful and realistic way using limited data during early stages. Since the project is developed as an academic exercise, collecting sufficient behavioral data for AI models may take time, requiring the use of hybrid rule-based and machine-learning approaches.

User reliability and no-shows also present a challenge. Designing a fair reputation system that discourages misuse without being overly strict requires careful balance. Additionally, managing workload alongside academic responsibilities and coordinating consistent development progress across the semester will require strong time management.

These challenges will be addressed through phased development, regular testing, supervisor feedback, and incremental improvements.

11. Project Success Criteria

The success of the PLAYERZWZ project will be measured by both technical performance and user experience. The project will be considered successful if users are able to find players more quickly, organize games with fewer cancellations, and experience smoother coordination compared to traditional methods.

From a technical standpoint, success includes stable application performance, accurate AI-based recommendations, and secure handling of user data. The AI matchmaking and reputation system should demonstrate logical and consistent behavior when tested with real users.

Academically, the project will be successful if it meets the learning objectives of the final year program, demonstrates proper application of software engineering and AI concepts, and is positively evaluated by the project advisor and examination committee.

12. Related Work / Literature Survey / Literature Review

Sports coordination and matchmaking systems have gained attention in recent years due to the increasing use of mobile applications for social and recreational activities. Previous studies show that digital platforms can significantly improve participation in sports by reducing coordination effort and increasing accessibility. However, many existing systems rely heavily on manual inputs and lack intelligent decision-making features.

Research in location-based services highlights the importance of proximity and availability in real-time matching applications. Similarly, studies in recommendation systems emphasize that personalized suggestions improve user engagement and satisfaction. Most existing sports-related applications either focus on venue booking or social networking but do not integrate AI-based matchmaking and trust evaluation.

PLAYERZWZ builds upon these ideas by combining location awareness, AI-assisted recommendations, and reputation scoring into a single platform. Unlike traditional solutions, the proposed system focuses on reducing waiting time and improving reliability, making it more suitable for spontaneous sports activities.

13. Project Plan / Project Schedule / Project Timetable / Project Calendar

Months 1–2:

- Requirement analysis and feasibility study
- Understanding user needs through surveys and observations
- Finalizing system architecture (mobile app, backend, AI modules)
- Selection of tools and technologies: Flutter, Firebase, Flask, Python, Scikit-learn

Months 3–4:

- Backend development using Flask
- Firebase Authentication and Firestore database setup
- User profile, game creation, and basic matchmaking APIs
- Location-based logic implementation

Months 5–6:

- Flutter mobile application development
- User registration, login, and profile management screens
- Game creation and game joining interfaces
- Integration of Google Maps for location services

Months 7–8:

- AI-assisted player matchmaking logic
- Development of rule-based + ML hybrid recommendation system
- Implementation of player rating and reliability scoring
- Availability mode and smart notifications

Months 9–10:

- No-show detection and misuse identification logic
- Testing AI recommendations with sample user data
- Performance optimization and bug fixing
- User experience improvements

Months 11–12:

- Final system testing and validation
- Deployment preparation and configuration
- Documentation and final report writing
- Final presentation, demo preparation, and viva readiness



14. References/Bibliography

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